Sean Luka Girgis

Data Engineer | Spark, Python, SQL, Data Quality & Production Pipelines

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across Python, SQL, PySpark/Spark, ETL/ELT pipelines, production data workflows, data ingestion, transformation, data quality, data modeling, data exports, testing, cloud data platforms, lakehouse-style patterns, documentation, and stakeholder-facing analytics. Strong background building and supporting reliable data pipelines, validating outputs, troubleshooting production jobs, optimizing SQL/reporting layers, and delivering trusted datasets for technical and business consumers. Best aligned to data engineering roles requiring Spark/PySpark, Python, SQL, production pipeline support, data quality, testing, cloud data platforms, data lake/lakehouse concepts, documentation, and collaboration; Azure, Microsoft SQL Server, Terraform, healthcare HL7/X12/FHIR, REST APIs, and Power BI are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable extraction, transformation, querying, analytics, and downstream consumption.
- Developed production data workflows for ingestion, cleansing, enrichment, aggregation, validation, and delivery to application, analytics, and business-facing reporting consumers.
- Implemented data quality and reconciliation practices across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, and AWS-backed reporting layers to improve completeness, consistency, and correctness of pipeline outputs.
- Supported production jobs by investigating source feeds, PySpark/Python transformation logic, SQL behavior, missing or late data, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Built and maintained data models, reporting datasets, metric definitions, validation checkpoints, and lineage-style documentation to improve interpretability and trust in analytics outputs.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Partnered with infrastructure, analytics, application, engineering, and business stakeholders to translate data requirements into reliable pipelines, dashboards, design notes, runbooks, and operational recommendations.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Led the FAST project, investigating real-user and synthetic monitoring source data to identify data quality issues, performance trends, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Partnered with engineering teams to explain tradeoffs, validate findings, and support practical performance improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data-processing pipelines.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and technical documentation used by application, middleware, infrastructure, and operations teams.
- Provided troubleshooting guidance, onboarding support, and knowledge sharing to improve adoption of monitoring standards and operational playbooks.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, regression thresholds, test findings, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, KornShell/ksh, C++, C, Java

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — Capacity Forecasting Engine

Built Python/PySpark forecasting workflows to process infrastructure telemetry, prepare forecasting datasets, validate pipeline outputs, and deliver capacity and bottleneck insights through dashboards.

Technologies: Python, Pandas, PySpark, SQL, Prophet, scikit-learn, Data Validation

ServiceCall AI RAG Demo

Built staged RAG proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, retrieval decisions, answer contract schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.

Technologies: Python, FastAPI, Docker, TF-IDF, Hybrid Retrieval, Pydantic, RAG

Enterprise APM Reporting Automation

Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance telemetry across large CA APM enterprise monitoring environments.

Technologies: KornShell/ksh, Perl, Unix/Linux, CA APM, SQL, Monitoring, Automation